1 $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$ **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

2 $5\sqrt{3}$ or $4\sqrt{3}$ or $3\sqrt{3}$ seen **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

3 $6\sqrt{2}$ or $5\sqrt{2}$ or $3\sqrt{2}$ seen **[M1]**

$8\sqrt{2}$ **[A1]**

[Total Marks: 2]

4 $5\sqrt{3}$ or $3\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

5 $\sqrt{25 \times 3} + \sqrt{16 \times 3} - \sqrt{9 \times 3} = 5\sqrt{3} + 4\sqrt{3} - 3\sqrt{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

6 (a) $3\sqrt{5}$ **[B1]**

(b) $2\sqrt{5}$ or $4\sqrt{5}$ seen **[M1]**

$6\sqrt{5}$ **[A1]**

[Total Marks: 3]

7 $\sqrt{75} = 5\sqrt{3}$ **[B1]**

$\sqrt{27} = 3\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[B1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 3]

8 $\sqrt{48} = 4\sqrt{3}$ **[B1]**

$\sqrt{12} = 2\sqrt{3}$ **[B1]**

$4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$ **[B1]**

[Total Marks: 3]

9 **(a)** M1 for $5\sqrt{3}$ or $2\sqrt{3}$

$7\sqrt{3}$ **[A1]**

(b) M1 for $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ or $\dfrac{14\sqrt{7}}{7}$

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

10 **(a)** $5\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$3\sqrt{3}$ **[A1]**

(b) $\dfrac{18\sqrt{6}}{6}$ **[M1]**

$3\sqrt{6}$ **[A1]**

[Total Marks: 4]

11 **(a)** $\sqrt{25 \times 3}$ or $\sqrt{25}\sqrt{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

(b) $6\sqrt{2} - 2\sqrt{2}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Total Marks: 4]

12 **(a)** **(i)** $3\sqrt{5} + 2\sqrt{5}$ **[M1]**

$5\sqrt{5}$ **[A1]**

(ii) Multiplies numerator and denominator by $\sqrt{3}$: $\dfrac{18\sqrt{3}}{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 4]

13 **(a)** $3\sqrt{5}$ **[B1]**

(b) $6\sqrt{2} + 5\sqrt{2} - 3\sqrt{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) $\dfrac{21\sqrt{7}}{\sqrt{7}\sqrt{7}}$ or $\dfrac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 5]

14 **(a)** $8 + 4\sqrt{3} - 2\sqrt{3} - 3$ **[M1]**

$5 + 2\sqrt{3}$ **[A1]**

(b) $\frac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 4]

15 (a) $9 - 3\sqrt{5} - 3\sqrt{5} + 5$ (or $9 - 6\sqrt{5} + 5$) **[M1]**

$14 - 6\sqrt{5}$ **[A1]**

(b) $\frac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

16 (a) $12 + 3\sqrt{5} - 8\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $\sqrt{7}$ to get $\frac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

17 (a) (a) $12 + 3\sqrt{5} - 8\sqrt{5} - 2(5)$ or $12 - 5\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) $\frac{15 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$ or $\frac{15\sqrt{3}}{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

[Total Marks: 4]

18 (a) $12 - 3\sqrt{2} + 4\sqrt{2} - 2$ **[M1]**

$10 + \sqrt{2}$ **[A1]**

(b) Multiply numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Total Marks: 5]

19 (a) M1 for $9 + 3\sqrt{2} + 3\sqrt{2} + 2$ (allow one numerical error in expansion)

$11 + 6\sqrt{2}$ **[A1]**

(b) M1 for multiplying numerator and denominator by $(3+\sqrt{5})$

M1 for denominator evaluated to 4 (from $3^2-(\sqrt{5})^2$)

$3+\sqrt{5}$ **[A1]**

[Total Marks: 5]

20 (a) $20-9$ or $4(5)-9$ **[M1]**

11 **[A1]**

(b) Multiplies numerator and denominator by $(3+\sqrt{2})$ **[M1]**

Denominator of $9-2$ or 7 seen **[M1]**

$6+2\sqrt{2}$ **[A1]**

[Total Marks: 5]

21 (a) $12+15\sqrt{2}-2\sqrt{2}-5$ **[M1]**

$7+13\sqrt{2}$ **[A1]**

(b) Multiplies numerator and denominator by $(4+\sqrt{3})$ **[M1]**

Denominator of $16-3$ or 13 **[M1]**

$8+2\sqrt{3}$ or $2(4+\sqrt{3})$ **[A1]**

[Total Marks: 5]

22 (a) $9-3\sqrt{5}-3\sqrt{5}+5$ **[M1]**

$14-6\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $(4+\sqrt{5})$ **[M1]**

$\dfrac{11(4+\sqrt{5})}{16-5}$ or denominator 11 seen **[M1]**

$4+\sqrt{5}$ **[A1]**

[Total Marks: 5]

23 (a) $\dfrac{15\sqrt{5}}{5}$ **[M1]**

$3\sqrt{5}$ **[A1]**

(b) $10-2\sqrt{3}+5\sqrt{3}-3$ **[M1]**

$7+3\sqrt{3}$ **[A2]**

[Total Marks: 5]

24 (a) $6-3\sqrt{5}+2\sqrt{5}-5$ **[M1]**

$1 - \sqrt{5}$ **[A1]**

(b) $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$\dfrac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 5]

25 (a) $15 + 5\sqrt{7} - 6\sqrt{7} - 2(7)$ **[M1]**

$1 - \sqrt{7}$ **[A1]**

(b) Multiplies numerator and denominator by $(5 + \sqrt{3})$: $\dfrac{22(5 + \sqrt{3})}{(5 - \sqrt{3})(5 + \sqrt{3})}$ **[M1]**

Denominator becomes $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Total Marks: 5]

26 (a) $20 + 8\sqrt{2} - 15\sqrt{2} - 6(2)$ **[M1]**

$8 - 7\sqrt{2}$ **[A1]**

(b) $\dfrac{11(4 + \sqrt{5})}{(4 - \sqrt{5})(4 + \sqrt{5})}$ **[M1]**

$\dfrac{11(4 + \sqrt{5})}{16 - 5}$ or $\dfrac{11(4 + \sqrt{5})}{11}$ **[M1]**

$4 + \sqrt{5}$ **[A1]**

(c) $3x\sqrt{3} - x\sqrt{3} = 18$ **[M1]**

$2x\sqrt{3} = 18$ or $x\sqrt{3} = 9$ **[M1]**

$x = \dfrac{9}{\sqrt{3}} = 3\sqrt{3}$ **[A1]**

[Total Marks: 8]

27 $\times \dfrac{3 + \sqrt{7}}{3 + \sqrt{7}}$ **[M1]**

Numerator $15 + 5\sqrt{7} + 3\sqrt{7} + 7$ or $22 + 8\sqrt{7}$ **[B1]**

Denominator $9 - 7$ or 2 **[B1]**

$11 + 4\sqrt{7}$ **[A1]**

[Total Marks: 4]

28 M1 for rearranging to group x terms, e.g., $x\sqrt{3} - x = 2 + \sqrt{12}$

M1 for factorising x, e.g., $x(\sqrt{3}-1) = 2+2\sqrt{3}$

M1 for multiplying by the conjugate to rationalise: $\dfrac{2+2\sqrt{3}}{\sqrt{3}-1} \times \dfrac{\sqrt{3}+1}{\sqrt{3}+1}$

M1 for correctly expanding the numerator to $8+4\sqrt{3}$ or evaluating the denominator to 2

$4+2\sqrt{3}$ **[A1]**

[Total Marks: 5]

29 Forms an equation for the area: $(a+b\sqrt{3})(2-\sqrt{3}) = 1+4\sqrt{3}$ **[M1]**

Expands the brackets: $2a - a\sqrt{3} + 2b\sqrt{3} - 3b = 1+4\sqrt{3}$ **[M1]**

Equates rational and irrational parts to form simultaneous equations: $2a-3b=1$ and $-a+2b=4$ **[M1]**

Attempts to solve the simultaneous equations (e.g. multiplies second equation by 2: $-2a+4b=8$ and adds to first to get $b=9$) **[M1]**

$a=14$, $b=9$ **[A1]**

[Total Marks: 5]

30 (a) Multiplies numerator and denominator by $(2+\sqrt{3})$ **[M1]**

Numerator expands to $10+5\sqrt{3}+2\sqrt{3}+3$ **[M1]**

Denominator is $4-3=1$ leading to final answer $13+7\sqrt{3}$ **[A1]**

(b) $x = \dfrac{14}{3-\sqrt{2}}$ **[M1]**

Multiplies numerator and denominator by $(3+\sqrt{2})$ to get $\dfrac{14(3+\sqrt{2})}{9-2}$ **[M1]**

$x = 6+2\sqrt{2}$ **[A1]**

[Total Marks: 6]

31 (a) Multiply numerator and denominator by $2+\sqrt{3}$ **[M1]**

Numerator becomes $(5+\sqrt{3})(2+\sqrt{3}) = 10+5\sqrt{3}+2\sqrt{3}+3$ (giving $13+7\sqrt{3}$) **[M1]**

$13+7\sqrt{3}$ **[A1]**

(b) Length $= \dfrac{12+4\sqrt{5}}{3+\sqrt{5}}$ **[M1]**

Multiply numerator and denominator by $3-\sqrt{5}$ **[M1]**

Numerator becomes $(12+4\sqrt{5})(3-\sqrt{5}) = 36-12\sqrt{5}+12\sqrt{5}-20 = 16$ **[M1]**

4 **[A1]**

[Total Marks: 7]

32 (a) Multiplies numerator and denominator by $(5+\sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3$ or 22 **[M1]**

$5 + \sqrt{3}$ **[A1]**

(b) $x(\sqrt{2} - 1) = \sqrt{8}$ or $x(\sqrt{2} - 1) = 2\sqrt{2}$ **[M1]**

$x = \dfrac{2\sqrt{2}}{\sqrt{2} - 1}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{2} + 1)$ **[M1]**

$4 + 2\sqrt{2}$ **[A1]**

[Total Marks: 7]

33 (a) $\dfrac{1}{2} \times (2 + \sqrt{3}) \times h = 5 + 3\sqrt{3}$ **[M1]**

$h = \dfrac{10 + 6\sqrt{3}}{2 + \sqrt{3}}$ **[M1]**

Multiply numerator and denominator by $(2 - \sqrt{3})$ **[M1]**

$h = \dfrac{20 - 10\sqrt{3} + 12\sqrt{3} - 18}{4 - 3} = 2 + 2\sqrt{3}$ **[A1]**

(b) $y^2 = (2 + \sqrt{3})^2 + (2 + 2\sqrt{3})^2$ **[M1]**

$y^2 = (4 + 4\sqrt{3} + 3) + (4 + 8\sqrt{3} + 12)$ **[M1]**

$23 + 12\sqrt{3}$ **[A1]**

[Total Marks: 7]

34 (a) $\dfrac{5 + 2\sqrt{3}}{2 + \sqrt{3}}$ **[M1]**

Multiplies numerator and denominator by $(2 - \sqrt{3})$ **[M1]**

$4 - \sqrt{3}$ **[A1]**

(b) $x\sqrt{5} - 2x = 2 + \sqrt{20}$ or $x(\sqrt{5} - 2) = \ldots$ **[M1]**

$x = \dfrac{2 + 2\sqrt{5}}{\sqrt{5} - 2}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{5} + 2)$ **[M1]**

$14 + 6\sqrt{5}$ **[A1]**

[Total Marks: 7]

35 (a) $\dfrac{14 - 2\sqrt{5}}{3 + \sqrt{5}}$ **[M1]**

Multiplies numerator and denominator by $(3 - \sqrt{5})$ **[M1]**

Numerator expands to $42 - 14\sqrt{5} - 6\sqrt{5} + 10$ (or $52 - 20\sqrt{5}$) **[M1]**

$13 - 5\sqrt{5}$ **[A1]**

(b) Uses Pythagoras' theorem: $(3 + \sqrt{5})^2 + (13 - 5\sqrt{5})^2$ **[M1]**

$14 + 6\sqrt{5}$ or $294 - 130\sqrt{5}$ seen **[M1]**

$308 - 124\sqrt{5}$ with no errors seen **[A1]**

[Total Marks: 7]

36 **(a)** Multiplies numerator and denominator by $(3 + \sqrt{3})$: $\dfrac{6(3+\sqrt{3})}{(3-\sqrt{3})(3+\sqrt{3})}$ **[M1]**

Expands denominator correctly: $9 - 3$ or 6 **[M1]**

$\dfrac{6(3+\sqrt{3})}{6}$ **[M1]**

$3 + \sqrt{3}$ **[A1]**

(b) $\dfrac{2\sqrt{6}+3\sqrt{6}}{\sqrt{3}}$ or $\dfrac{5\sqrt{6}}{\sqrt{3}}$ or $5\sqrt{2}$ **[M1]**

$\dfrac{10\sqrt{2}}{2}$ or $5\sqrt{2}$ **[M1]**

0 (or $0\sqrt{2} + 0\sqrt{3}$) **[A1]**

[Total Marks: 7]

37 **(a)** $\dfrac{1}{2}(\sqrt{7}-1)(\sqrt{7}+2)$ **[M1]**

Expands brackets correctly: $7 + 2\sqrt{7} - \sqrt{7} - 2 = 5 + \sqrt{7}$ **[M1]**

$\dfrac{5+\sqrt{7}}{2}$ **[A1]**

(b) $AC^2 = (\sqrt{7}-1)^2 + (\sqrt{7}+2)^2$ **[M1]**

$(\sqrt{7}-1)^2 = 8 - 2\sqrt{7}$ and $(\sqrt{7}+2)^2 = 11 + 4\sqrt{7}$ **[M1]**

$AC^2 = 19 + 2\sqrt{7}$ **[A1]**

$\sqrt{19 + 2\sqrt{7}}$ **[A1]**

[Total Marks: 7]

38 **(a)** Area $= \dfrac{1}{2} \times$ base $\times$ height $\implies 5 + 2\sqrt{2} = \dfrac{1}{2} \times (3 + \sqrt{2}) \times h$ **[M1]**

$h = \dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ **[A1]**

(b) Multiply numerator and denominator by $(3 - \sqrt{2})$: $\dfrac{2(5+2\sqrt{2})(3-\sqrt{2})}{(3+\sqrt{2})(3-\sqrt{2})}$ **[M1]**

Denominator: $3^2 - (\sqrt{2})^2 = 9 - 2 = 7$ **[B1]**

Numerator: $2(15 - 5\sqrt{2} + 6\sqrt{2} - 4) = 2(11 + \sqrt{2}) = 22 + 2\sqrt{2}$ **[M1]**

Height $= \dfrac{22+2\sqrt{2}}{7}$ **[A1]**

(c) $\dfrac{1(\sqrt{7}+2)}{(\sqrt{7}-2)(\sqrt{7}+2)}$ **[M1]**

$\dfrac{\sqrt{7}+2}{7-4} = \dfrac{\sqrt{7}+2}{3}$ **[A1]**

[Total Marks: 8]

39 **(a)** $\dfrac{1}{2} \times (\sqrt{6}-\sqrt{2})(\sqrt{6}+\sqrt{2})$ **[M1]**

$\dfrac{1}{2}(6-2)$ **[M1]**

2 **[A1]**

(b) $(\text{Hypotenuse})^2 = (\sqrt{6}-\sqrt{2})^2 + (\sqrt{6}+\sqrt{2})^2$ **[M1]**

$(6-2\sqrt{12}+2) + (6+2\sqrt{12}+2)$ or $8-2\sqrt{12}+8+2\sqrt{12}$ **[M1]**

$(\text{Hypotenuse})^2 = 16$ **[A1]**

$\text{Hypotenuse} = \sqrt{16} = 4$ **[A1]**

(c) $\text{Area scale factor} = \dfrac{18}{2} = 9$ **[M1]**

$\text{Length scale factor} = \sqrt{9} = 3$ **[M1]**

$4 \times 3 = 12$ **[A1]**

[Total Marks: 10]